

Faculty of Science, Engineering and Built Environment

SEE406 Power System Analysis

Deakin University Unit Guide

Trimester 2, 2025

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Welcome

Power system is comprised of several complex elements and appropriate analytical and modelling tools are required to investigate their interaction with each other as well as their impact on overall system stability. This unit covers electrical power system modelling, symmetrical components, load flow analysis using various types of load flow methods, balanced and unbalanced fault calculations, power system stability and control. Students will be able to design a power substation using industry standard software and perform load flow analysis to evaluate the network performance. The skills developed through this unit will help students implement advanced technologies for ensuring effective operation of the electricity networks in the presence of renewable energy sources.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Ameen Gargoom
Campus: Geelong Waurn Ponds, CADET Building, Level 3
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Other members of the team and how to contact them

You will be advised of details of any other staff members involved in teaching this unit via the unit site.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- the lecture content and recordings
- detailed sessions on the software.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- updated tutorial problems
- updated studio sessions with more exercises.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Apply technical knowledge to analyse power system stability on a system.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy GLO4: Critical thinking GLO5: Problem solving
ULO2	Apply technical knowledge to analyse load-flow and fault calculations in power systems.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO4: Critical thinking GLO5: Problem solving GLO6: Self-management
ULO3	Demonstrate a working knowledge of power system analysis relevant to engineers involved in both operations and planning.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO4: Critical thinking GLO5: Problem solving GLO6: Self-management
ULO4	Demonstrate a working knowledge of relevant industrial safety standards.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO4: Critical thinking GLO5: Problem solving
ULO5	Recommend and apply the correct safety protocols for power systems.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO4: Critical thinking
ULO6	Communicate with engineering teams and the community at large to investigate and present different power systems operation.	GLO2: Communication GLO5: Problem solving

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 40% in the combined topic test and quiz marks.

Brief summary of the hurdle requirement	Rationale
To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 40% in the combined topic test and quiz marks.	This unit requires a pass in the topic test to ensure that each student can show their independent learning which covers ULO 1,2,3,4.

Summative assessment (tasks that will be graded or marked)
NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.
Summative assessment task 1

Title	Assignment 1 (individual) - Online Quiz
Brief description of assessment task	This is an individual quiz assignment based on topics covered in Weeks 1 - 5. Students will be expected to solve problems and answer conceptual questions based on power flow analysis and economic dispatch.
Detail of student output	A 90-minute online quiz. Students will be asked to answer online questions based on the covered topics in Weeks 1 -5. Students will demonstrate problem solving skills as well as technical knowledge on different aspects of power system analysis.
Grading and weighting (% total mark for unit)	20% (Marked)
This task assesses your achievement of these Unit Learning Outcome(s)	By successfully completing this assignment students will achieve the following ULOs: ULO1: Apply technical knowledge to understand and analyse power system operation. ULO2: Apply technical knowledge to analyse load-flow and fault calculations in power systems. ULO3: Demonstrate a working knowledge of power system analysis.
This task assesses your achievement of these Graduate Learning Outcome(s)	By successfully completing this assignment students will demonstrate discipline-specific knowledge, show problem solving skill with critical thinking and work or learn independently which indicates their achievement of GLO1, GLO2, GLO4, GLO5.
How and when you will receive feedback on your work	Students will be given adequate feedback and support during the trimester and overall unit feedback at the end of the trimester during the revision of the unit.
When and how to submit your work	**Monday, 4 August 2025 (Week 5)** via the unit site.

Summative assessment task 2

Title	Assignment 2 (individual) - Problem solving and software tasks
Brief description of assessment task	Students are required to submit a written report divided into two parts. Part 1 includes three tasks, and Part 2 includes two tasks. These tasks will involve solving mathematical problems and analysing outcomes from software-based studio activities and present finding and the analysis in the form of a professionally written report.

Detail of student output	This is an individual assessment task. Students are required to submit the working solutions/simulation output with analysis report. In addition, students are required to present the detailed calculations and model findings in the report. Students are expected to demonstrate their problem-solving skills as well as their ability to reflect critically on the findings from studio activities Further details will be provided in the unit site.
Grading and weighting (% total mark for unit)	30% (Part 1 (18%), Part 2 (12%) (Graded)
This task assesses your achievement of these Unit Learning Outcome(s)	Upon the successful completion of this assignment students will achieve the following ULOs: ULO1: Apply technical knowledge to understand and analyse power system operation. ULO2: Apply technical knowledge to analyse load flow and fault calculations. ULO3: Demonstrate working knowledge of power system analysis.
This task assesses your achievement of these Graduate Learning Outcome(s)	By successfully completing this assignment students will demonstrate discipline-specific knowledge, show problem solving skill with critical thinking, communicate effectively and work or learn independently or in a team which indicates their achievement of GLO1, GLO3, GLO4 & GLO5.
How and when you will receive feedback on your work	Students will receive feedback through the unit site prior to the submission of the assignment. Feedback and/or solutions will be available after the submission date.
When and how to submit your work	**Part 1** due by 8pm (AEST) on Monday 18 August 2025 (Week 6) **Part 2** due by 8pm (AEST) on Friday 12 September (Week 9) Online submissions via the unit site.

Summative assessment task 3

Title	**Assignment 3 (individual) – Topic test**
Brief description of assessment task	Students will be required to submit written responses to problems and conceptual questions based on different aspects of power system analysis including symmetrical and unsymmetrical faults, power system stability and control.
Detail of student output	In this assessment task, students are required to provide answers to number of problem based and conceptual questions individually. The tests will require both written responses and mathematical calculations. Students will apply their understanding of power systems analysies and design solutions to ensure a stable and efficient operation of the power system network.
Grading and weighting (% total mark for unit)	20% (Marked)
This task assesses your achievement of these Unit Learning Outcome(s)	By successfully completing this assignment students will achieve the following ULOs: ULO1: Apply technical knowledge to understand and analyse power system operation. ULO3: Demonstrate a working knowledge of power system analysis. ULO4: Demonstrate working knowledge of safety standard.
This task assesses your achievement of these Graduate Learning Outcome(s)	By successfully completing this assignment students will demonstrate discipline-specific knowledge, show problem solving skill with critical thinking and work or learn independently which indicates their achievement of GLO1, GLO2, GLO4, GLO5.
How and when you will receive feedback on your work	Students will be given adequate feedback and support during the trimester and overall unit feedback at the end of the trimester during the revision of the unit.
When and how to submit your work	**Thursday 25 September 2025 (Week 11)** by 8pm AEST, via the unit site.

Summative assessment task 4

Title	**Assignment 4 (individual) - Project report**
Brief description of assessment task	This is project work. Students will be required to develop a software model using PSS/Sincal to design solutions for an electrical substation considering the network aspects and analyse the findings.
Detail of student output	This is an individual assessment task, however students can work together to design and develop the model. Students can form teams to start their design, before analysing the design individually. They are also required to perform load flow or power flow simulation and include findings in the project report. Students will present the design of the electrical substation, incorporate safe design considerations and reflect critically on the findings of the short circuit analysis. Students are expected to demonstrate their ability to use industry standard power system software and develop and analyse simulation models. Further details will be provided in the unit site.
Grading and weighting (% total mark for unit)	30% (Graded)
This task assesses your achievement of these Unit Learning Outcome(s)	By successfully completing this assignment students will achieve the following ULOs: ULO2: Apply technical knowledge to analyse load-flow and fault calculations in power systems. ULO3: Demonstrate a working knowledge of power system analysis. ULO4: Demonstrate working knowledge of safety standards. ULO5: Recommend and apply correct safety protocols ULO6: Communicate with engineering team to investigate power systems.
This task assesses your achievement of these Graduate Learning Outcome(s)	Upon successfully completing this assignment students will demonstrate discipline specific knowledge, apply critical thinking while using digital literacy in simulation work. They will also demonstrate their problem solving skills in a team environment by communicating effectively with diverse community and therefore achieve GLO1, GLO2, GLO3, GLO4, GLO5, GLO6.
How and when you will receive feedback on your work	The lecturer/tutor will facilitate the project activities during the project work and will provide collective feedback on discussion session.
When and how to submit your work	**Friday 3 October 2025 (end of Week 12) by 8pm (AEST)** via the unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 1 hour seminar per week, 1 x 2 hour practical experience (studio) per week.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 2 hour practical experience (studio) per week.

Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

For campus students learning activities include watching recorded content and attending on-campus seminars and studios. For online students learning activities include watching recorded content and attending online studio workshops. Students are also expected to spend time reading the practical resources provided and completing self-directed problem-based exercises and assessment tasks.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SEE406 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

There is no prescribed textbook for this unit.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Recommended textbook:

- Glover and Overbye, 2023, Power System Analysis and Design, 7th edition, (or 6th SI Edition), Cengage Learning US.

The following textbook is also recommended. Although this text should be quite useful, it is not essential for the successful completion of SEE406.

- Power System Analysis, Hadi Saadat, 3rd Edition. ISBN: 978-0-9845438-0-9

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 2 teaching period begins	Monday 7 July 2025
Census Date	Friday 15 August 2025
Intra-trimester break	Monday 11 August - Sunday 17 August 2025
Trimester 2 teaching period ends	Friday 26 September 2025
Study period (end-of-unit assessment/examination preparation period)	Monday 29 September - Friday 3 October 2025
End-of-unit assessment/examination period	Monday 6 October - Friday 17 October 2025
Inter-trimester break (the period between trimesters)	Monday 20 October - Friday 31 October 2025
Unit results released	Friday 31 October 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Assessment due date
1	7 July 2025	Power System Overview	
2	14 July 2025	Power System Overview	
3	21 July 2025	Power flow analysis	
4	28 July 2025	Power flow analysis	
5^	4 August 2025	Economic dispatch	**Assessment 1: Online quiz** due 4 August
6	18 August 2025	Balanced fault analysis	**Assessment 2: Problem solving and software task (Part 1)** due 18 August
7	25 August 2025	Symmetrical Components	
8	1 September 2025	Unbalanced fault analysis	
9	8 September 2025	Power System stability	**Assessment 2: Problem solving and software task (Part 2)** due 12 September
10	15 September 2025	Power system control	
11#	22 September 2025	Revision	**Assessment 3: Topic test** due 25 September
12	29 September 2025		Study period **Assessment 4: Project report** due 3 October
13-14	6 October 2025		End-of-unit assessment and exam period

^ Intra-trimester break: **Monday 11 August – Sunday 17 August 2025**
AFL Grand Final Eve public holiday: **Friday 26 September 2025** - University closed